Paper Walkthrough • arXiv 2611.05163

RoboTryOn

Synthesizing Dexterous Robot Demonstrations from Human Videos via Virtual Try-On

Like virtual try-on: keep the human motion, swap human hands for robot hands,**synthesizing trainable dexterous-hand robot demos from human videos**

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Overview: human video → robot video → action labels → new task skills

2.1% → 60.0%
Avg. over 7 new tasks without robot demos, two robots
(teleop only → teleop + synthetic)

45.9%
Avg. bimanual task success,
though all teleop demos are single-arm

≈ 1/3 cost
Human videos alone match 50 teleop demos/task
(1.87 vs 5.26 person-hours)

One-sentence summary

RoboTryOn treats human-to-robot transfer as **virtual try-on**: autoregressive keyframe translation + hand-skeleton-guided video interpolation produce target robot videos, then an **inverse dynamics model (IDM)** reads joint actions from them; the generators train on robot data only, with no paired human–robot data.

① Keyframe translation

An image model inpaints masked human keyframes

into robot keyframes

② Video synthesis

A video model interpolates between keyframes

Hand-skeleton stream guides motion

③ Action pseudo-labels

IDM predicts joint actions from generated video

Wrist corrected via hand reconstruction + IK

④ Policy training

Synthetic demos share the teleop format

Fine-tune π0.5 / RLDX-1-Img

Problem: cheap human video; dexterous hands face visual/action gaps

Visual gap

Training frames show human hands and arms; deployment shows the robot.

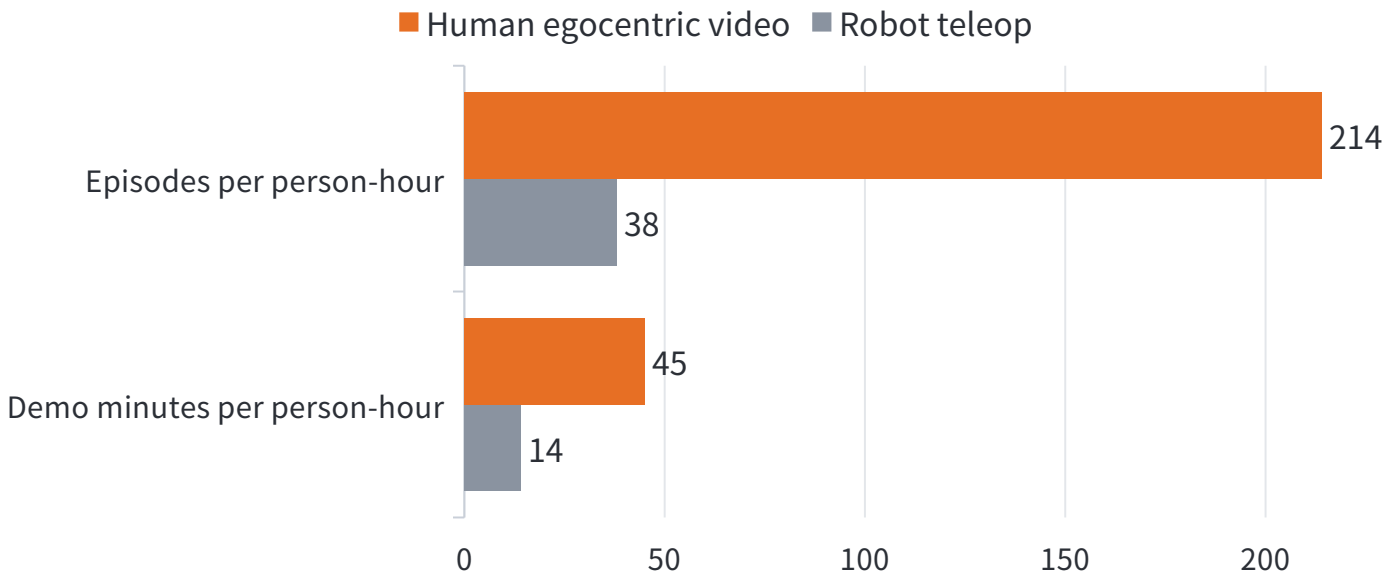
Prior work: EgoMimic masks the arm, Masquerade overlays a rendered robot — both must apply the same edit to live camera frames at deployment.

Action gap

Hand reconstruction is itself inaccurate; human and robot hands differ in morphology/kinematics, so no exact retargeting map exists.

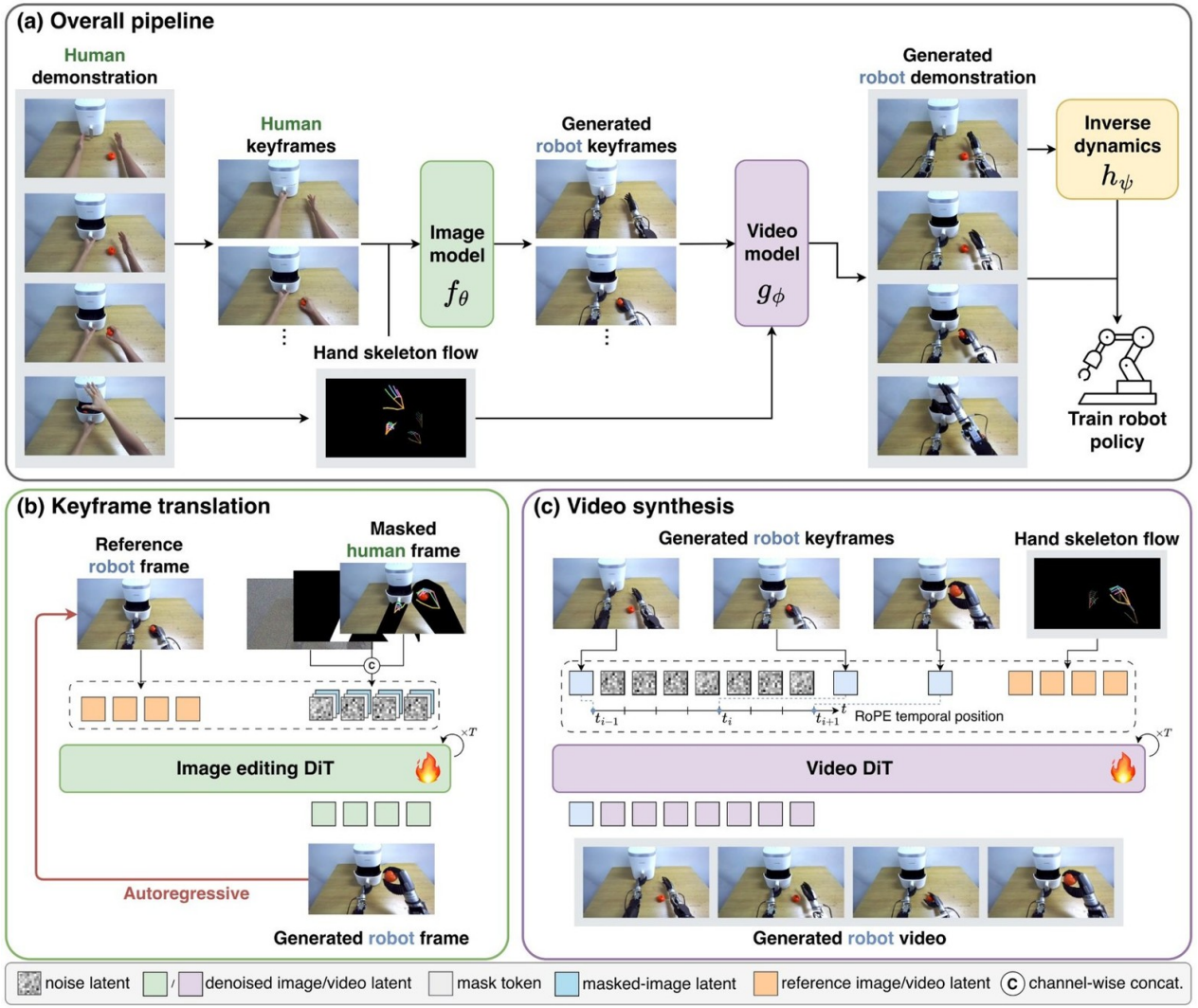
Even with perfect shape mapping, grip force remains unknown.

Why it matters: 5–6× collection cost gap (Table 10)



- Teleop: 2 people/session; needs robot + teleop system + trained operator; costly for bimanual
- Human video: 1 person + fixed camera, no robot time, no extra bimanual cost
- Goal: learn new tasks from human videos alone

Core idea: like virtual try-on, “keep the motion, swap the performer”



Three-stage pipeline

(a) Human keyframes → image model f_θ → robot keyframes → video model g_ϕ (+ hand-skeleton stream) → robot video → IDM h_ψ → actions

(b) Keyframe translation inpaints masked human frames using the previous robot keyframe as reference; each result autoregressively becomes the next reference

(c) Video synthesis places keyframes at RoPE temporal positions, feeds the skeleton stream as in-context tokens, and generates in-between frames

Robot-only training: robot frames are likewise masked and overlaid with skeletons; at inference, simply swap in human-frame masks and skeletons

Key design: embodiment-agnostic cues avoid pairing & retargeting error

Mask + skeleton conditioning

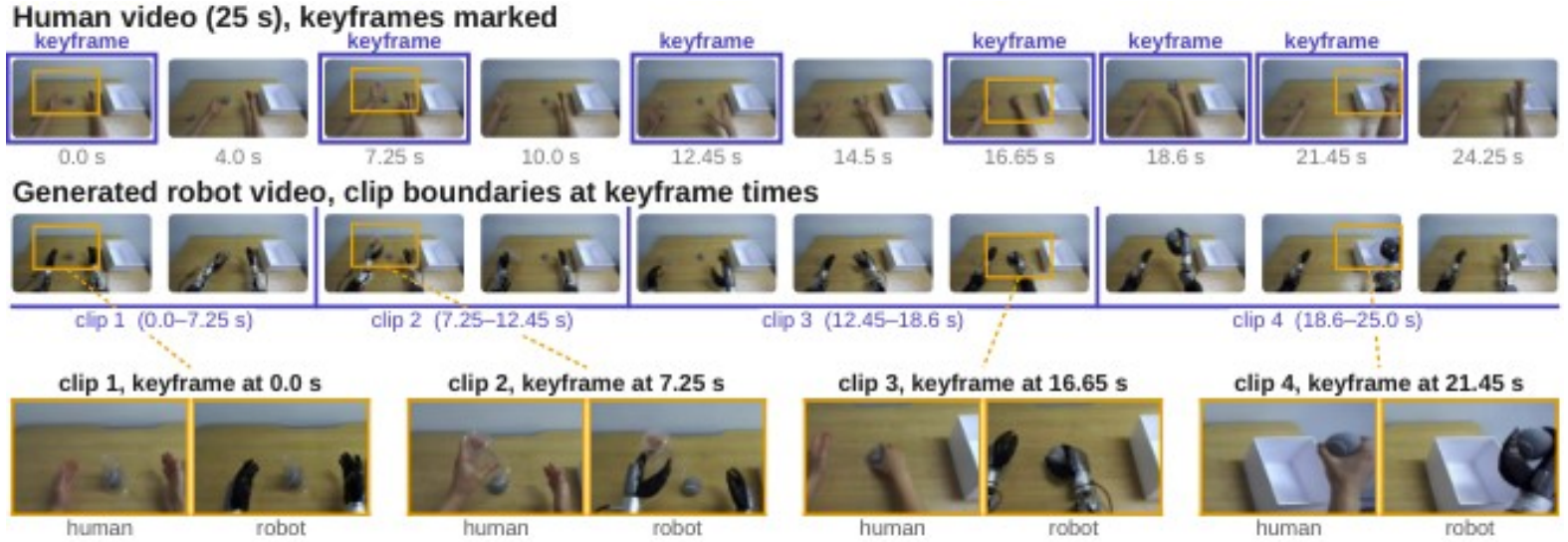
- Masks are dilated and smoothed so contours don't reveal whether a human or robot is hidden
- A 17-point hand skeleton unifies human / 6-DoF / 20-DoF robot hands
- Training-time skeleton jitter (⩽ half a phalanx), full dropout p=0.2, per-point dropout p=0.2 → coarse guidance only

Keyframes & long videos

- Keyframes: local minima of both wrists' speed (no hand blur) with above-threshold displacement; first and last always kept
- ~4 keyframes for PnP, 6–7 for long tasks
- ⩽145 frames per segment (~7.25 s); adjacent segments share boundary keyframes, enabling 10–25 s demos

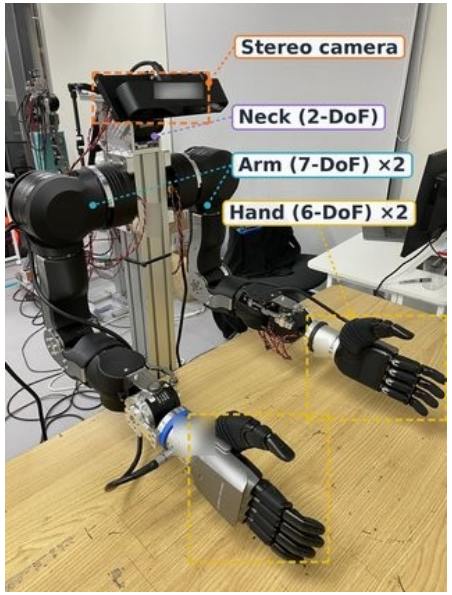
IDM pseudo-action labeling

- Depth Anything V2 backbone + flow-matching action head
- Two frames 1 s apart → 20-step joint action chunk; sliding-window averaging + S-G smoothing
- Wrist translation corrected via hand reconstruction + IK; finger joints keep IDM predictions



Source: § 3, Appendix B.2, Figure 14

Setup: two bimanual dexterous humanoids, 11 real-world tabletop tasks



OpenArm
7-DoF arm + 6-DoF Inspire hand

RB-Y1
7-DoF arm + 20-DoF WujiHand2

- Head-mounted stereo camera; fixed neck/base; human videos recorded from the robot's head-camera pose
- Policy: π0.5 fine-tuned 30k steps; 200 teleop / 550 synthetic / 750 total demos
- 28 trials per task, randomized object positions

4 in-domain PnP tasks

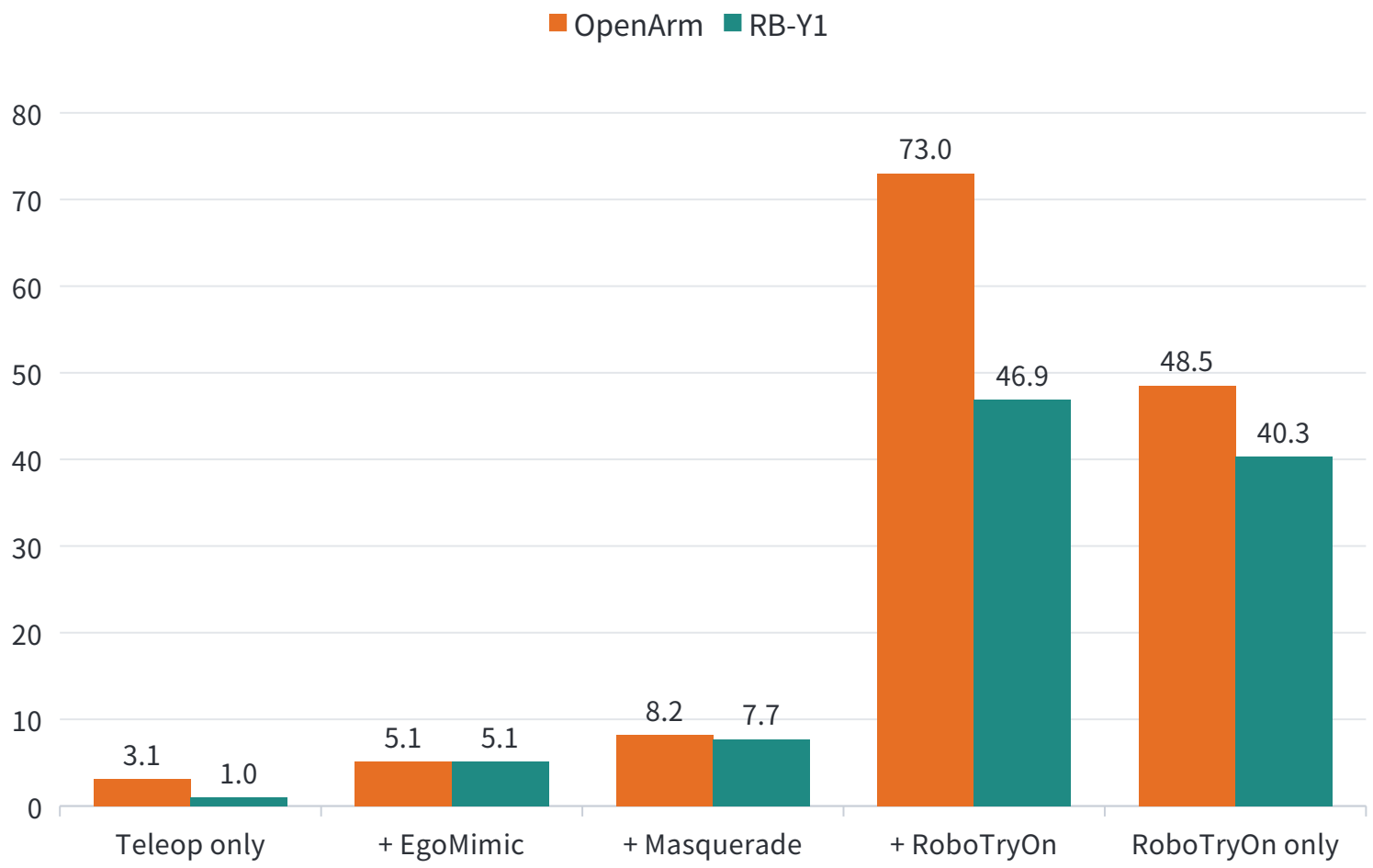
Bottle / Doll / Box / Ball
50 teleop demos + 50 human videos each

7 out-of-domain tasks (human video only)

New object: Coffee Pot • New goal: Bottle to Shelf
New skills: Pouring / Brushing
Bimanual: Box Convey / Air Fryer / Ball in Cup

Results: baselines nearly all fail on new tasks; RoboTryOn succeeds

Avg. success on 7 out-of-domain tasks (%)



Avg. on 4 in-domain tasks (%)

	OpenArm	RB-Y1
Teleop only	47.3	69.6
+ Masquerade	67.9	63.4
+ RoboTryOn	**75.9**	**79.5**

< 10%

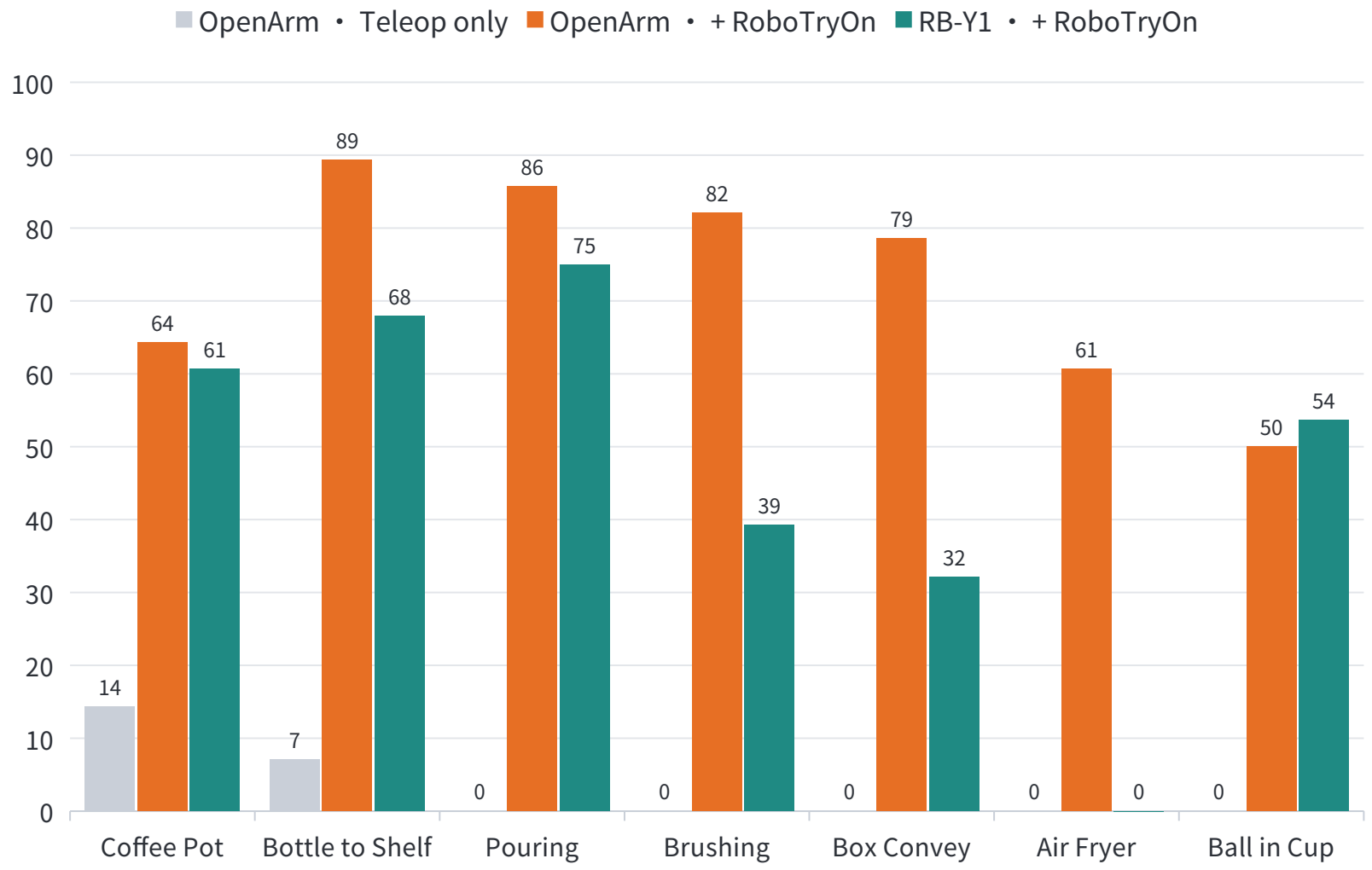
Avg. out-of-domain success of the two baselines
(retargeted actions + test-time frame editing)

40–49%

Out-of-domain success of a policy trained without teleop
(RoboTryOn synthetic data only)

Source: Table 1 / Table 11 (success %, 28 trials per task)

Per task: new objects, goals, skills, even bimanual coordination are learned



Key takeaways

Single-arm teleop → bimanual policy: 78.6 / 60.7 / 50.0 on OpenArm's three bimanual tasks

RB-Y1 Air Fryer = 0%: fingers must slip between handle and body, beyond teleop data and IDM label precision; long 20-DoF fingers tend to push the fryer away when closing it

Bimanual scoring: 0.5 points per completed hand, partial credit included

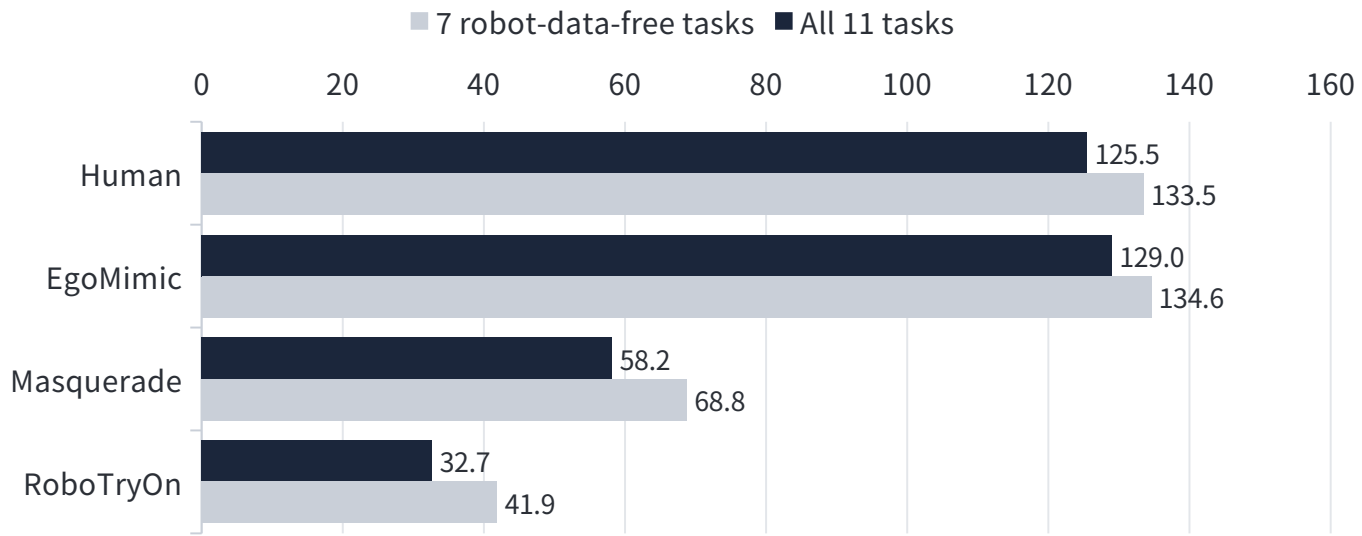
Statistical caveat: 28 trials per task; per-task values vary by about ±9 points

Source: Table 11 (out-of-domain tasks, success %)

Visual gap: synthesis closest to real robot; no deploy-time edits

Raw human video | EgoMimic (masking) | Masquerade (overlay) | RoboTryOn (synthesis) | Real teleop

Fréchet SigLIP distance to real robot frames (lower = closer)



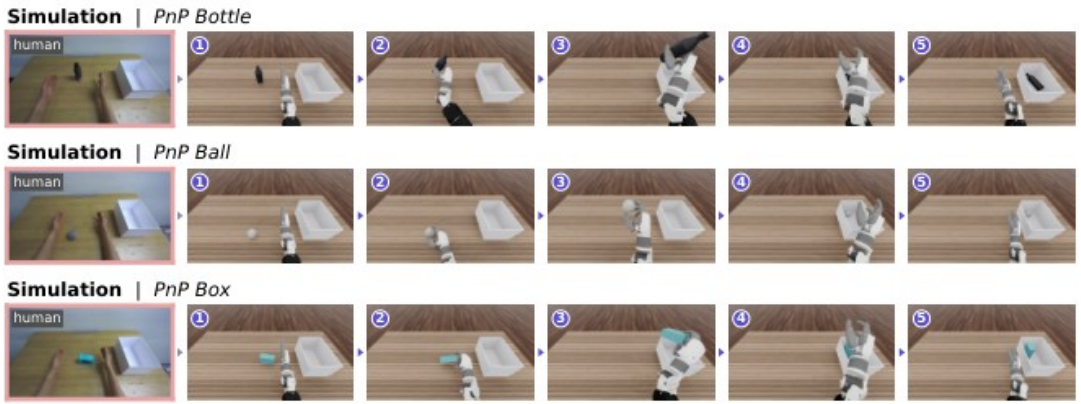
125.5 → 32.7

RoboTryOn cuts distance to real robot frames to ~1/4;
overlay roughly halves it, masking barely changes it

- Features from π0.5's SigLIP vision encoder
- Still only 41.9 on the 7 robot-data-free tasks → not reproducing the training set
- Baselines close the gap only by applying the same mask/overlay to every frame at deployment

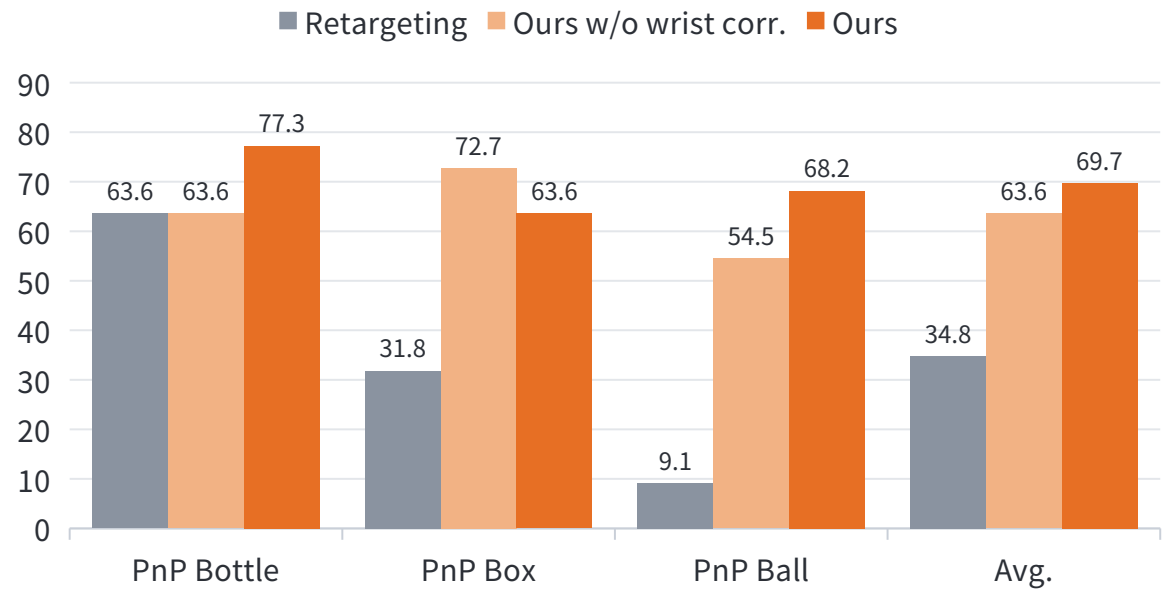
Source: Figure 3(a), Table 7 (Fréchet SigLIP distance)

Action gap: IDM- “read” actions more executable than retargeting



Isaac Lab replica: human video → open-loop replay (grasp, lift, place)

Sim open-loop replay success (%), 22 per task



69.7% vs 34.8%

Replay success: IDM vs retargeted labels
gap arises mainly at grasping

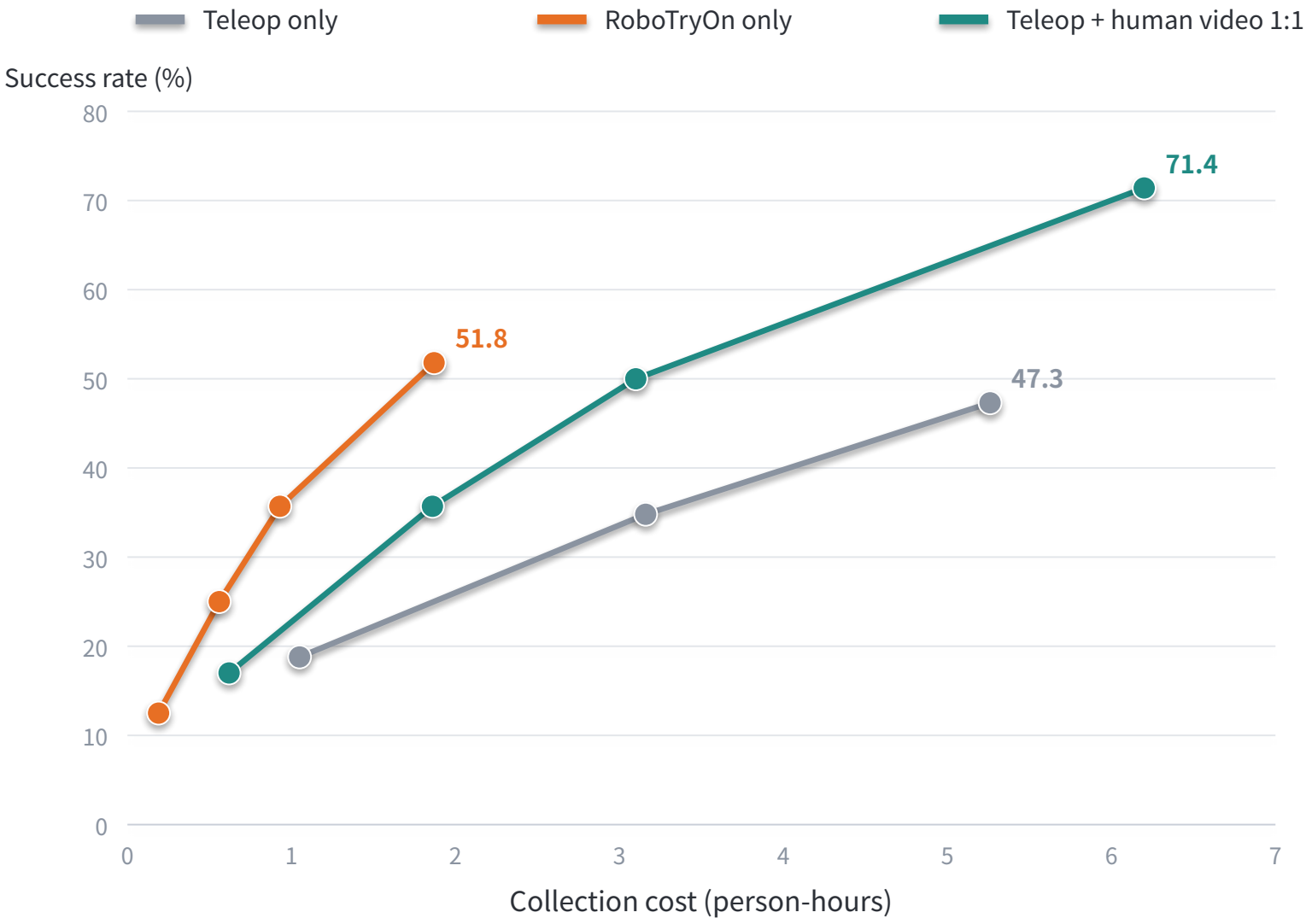
0.53 vs 0.17 / 0.14

Policy-gradient cosine similarity with teleop data
(Ours vs EgoMimic / Masquerade)

- Retargeting: hands reach the object, but reconstruction error and hand-shape mismatch cause weak grasps
- Retargeted data has slightly negative gradient cosine on the action expert → conflicts with teleop data
- Real rollouts: baseline OOD hand motions drift from the teleop distribution; Ours stays close

Source: Figure 4, Figure 5, Table 8, Figure 9

Cost: each human video slightly weaker, far better per person-hour



51.8% @ 1.87h

Human video only (100/task)
⩾ teleop (50/task): 47.3% @ 5.26h

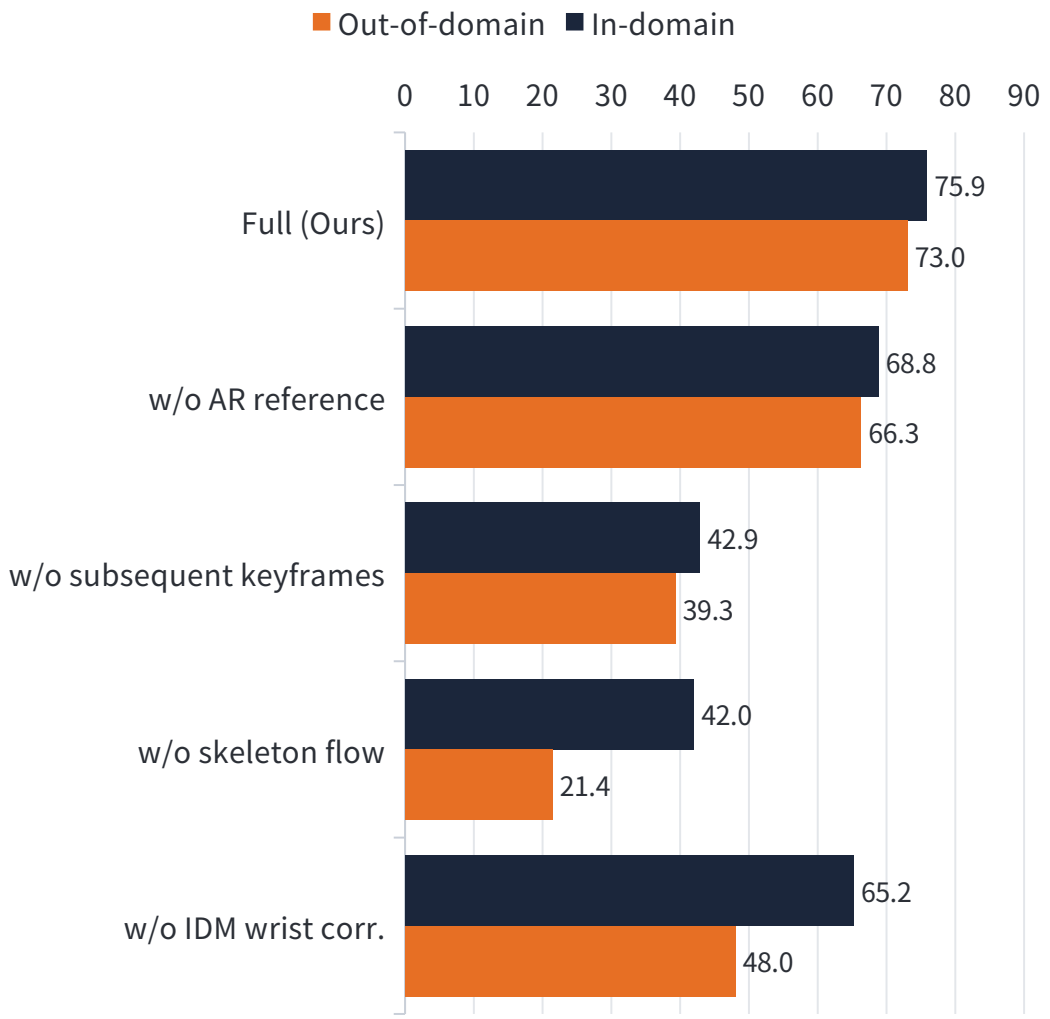
71.4%

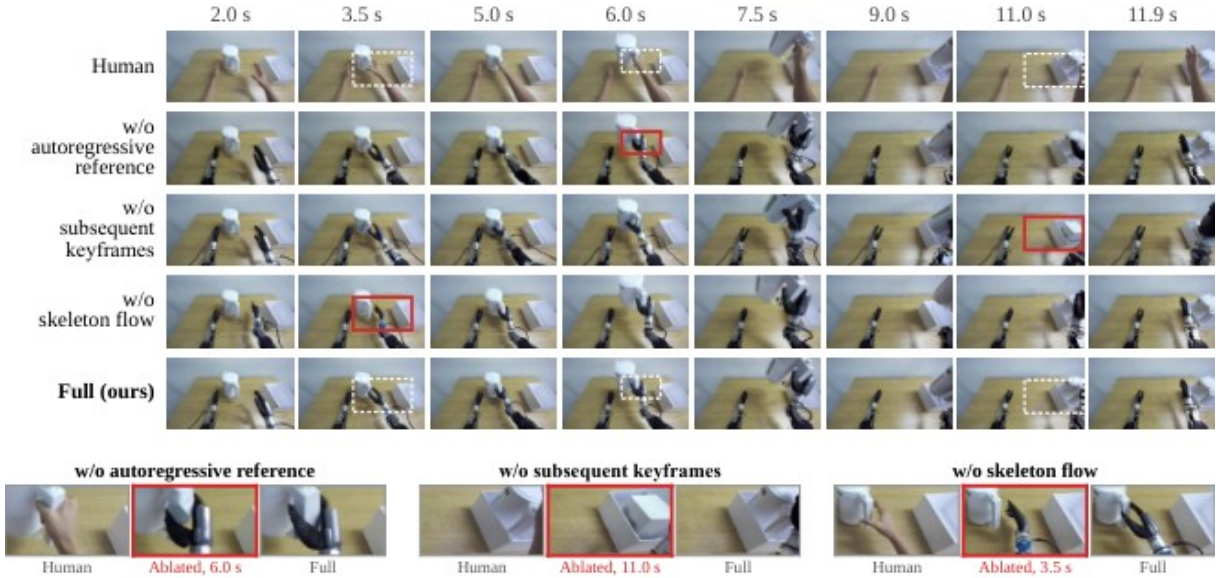
Teleop + human video 1:1 (50 each/task)
6.20 person-hours total

- At 50/task each: human video 35.7% vs teleop 47.3%
- Person-hours exclude a one-time few hours of teleop per robot (to train generators/IDM) and GPU synthesis compute

Source: Figure 6, Table 9 (4 PnP tasks, 112 trials)

Ablations: hand-skeleton stream matters most; all parts essential





w/o skeleton stream: no motion guidance for in-between frames, which stop following the demonstrator (OOD 21.4%)

w/o subsequent keyframes: motion preserved but objects drift; the coffee pot becomes a box at 11 s

w/o autoregressive reference: inconsistent grasp configurations across keyframes cause hand jitter

w/o wrist correction: larger out-of-domain drop (73.0→48.0); new motions exceed IDM's training distribution

Source: Table 3(a), Figure 11 (OpenArm, π0.5)

Two findings: policies inherit demo speed; gains hold for other VLAs

Speed (resynthesis after skeleton retiming)

Data	Speed	ID	OOD	Time (s)
Teleop only	1×	47.3	3.1	14.2
Ours (orig. speed)	1.25×	75.9	73.0	11.9
Ours （retimed）	**1×**	**85.7**	**70.4**	**13.8**
Ours （retimed）	2×	57.1	42.9	7.4

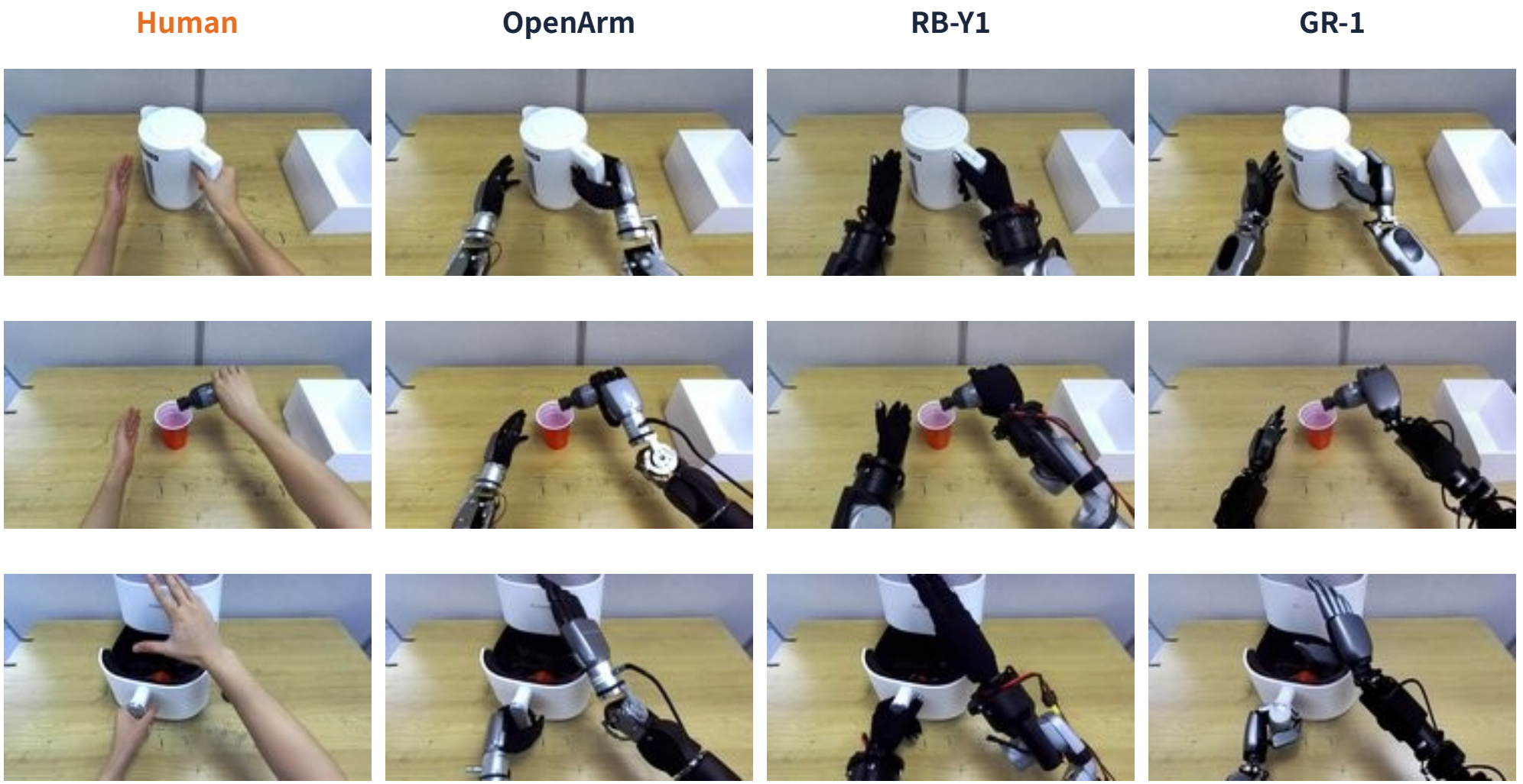
- Human demos run at ~1.25× teleop speed
- Retiming the skeleton stream to 1× per phase: in-domain 75.9 → 85.7, out-of-domain roughly flat
- 2× is fastest but success drops markedly → policies inherit training-demo speed

Policy backbones: π0.5 and RLDX-1-Img

Policy	ID	OOD
π0.5 （Teleop only）	47.3	3.1
+ RoboTryOn	**75.9**	**73.0**
RLDX-1-Img （Teleop only）	53.6	0.0
+ RoboTryOn	**60.7**	**40.3**

- RLDX-1-Img uses a GR00T N1-style separate action head, unlike π0.5's architecture
- Out-of-domain 0.0 → 40.3: synthetic-data gains are not tied to a specific VLA

One model, many robots: same human video, only the performer changes

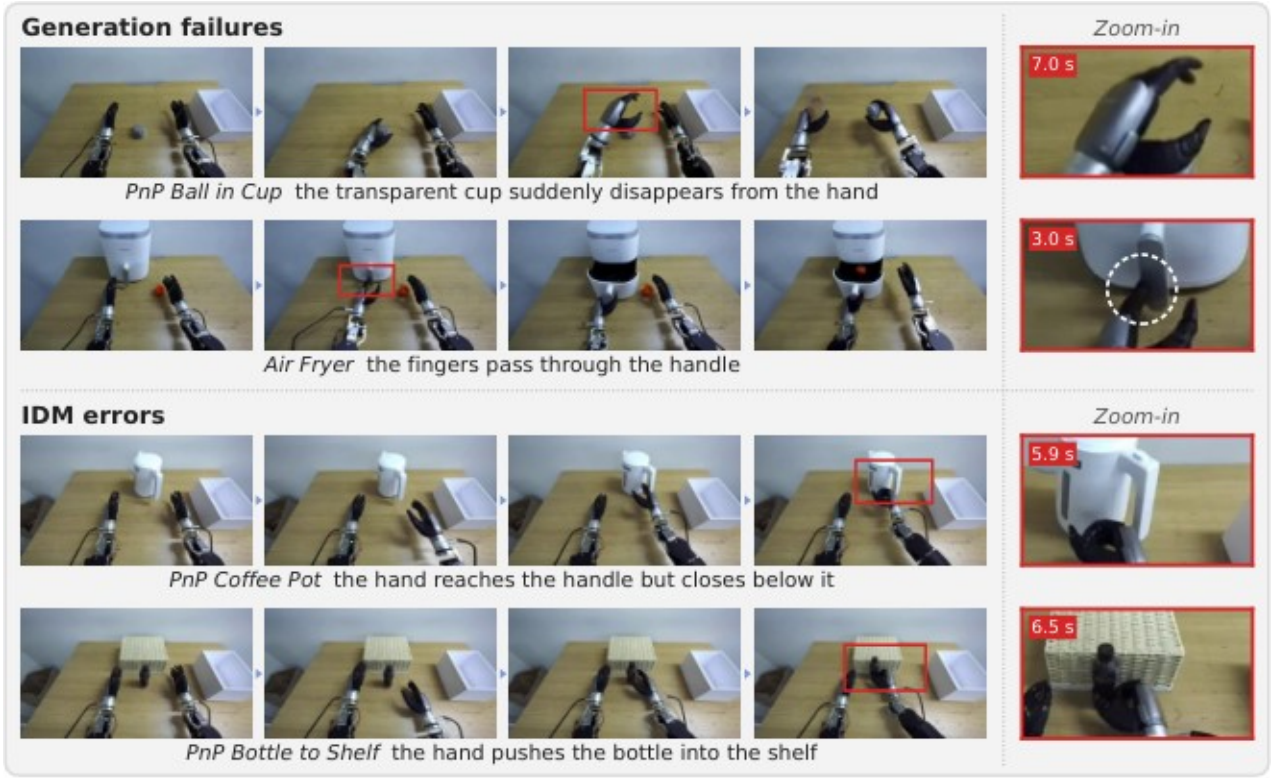


Same image model + video model + IDM

Training data: OpenArm 3.79 h, RB-Y1 2.53 h, GR-1 2.50 h of teleop

GR-1: video translation only, no real-robot policy evaluation

Limitations: generation/label errors; controlled views & robot priors



Still needs robot data

Each new robot needs a few hours of teleop (any task) to train the generators and IDM; the training corpus also includes teleop demos of the 4 in-domain tasks

Fixed viewpoint

Human videos recorded from the robot head-camera pose with fixed neck/base; in-the-wild moving-view videos are out of distribution

Pseudo-label accuracy

Cannot express force; not precise enough for fine tasks like tight-fit insertion; wrist correction depends on camera calibration

Trading compute for labor

A 15 s demo takes ~695 s (≈11.6 min) on a single H100

Executability

Can retime, but cannot fix grasps the robot can't reach or its hand can't form

Summary: swap appearance, keep motion, read actions

Swap appearance

Generative “virtual try-on”
mask + skeleton interface, robot-only training, no pairing; raw camera frames at deployment

Keep motion

Autoregressive keyframes + skeleton-guided interpolation
consistent scene, objects and demonstrator motion; supports 10–25 s demos

Read actions

IDM reads joint actions from generated video + wrist IK correction
more executable than retargeting; gradients align with teleop

Takeaways

- For high-DoF dexterous hands, “generated robot video + IDM labels” bridges the human–robot gap better than “masking/overlay + retargeting”
- On two real robots: new tasks 2.1% → 60.0%, bimanual 45.9%; matches teleop at ~1/3 the collection cost
- Key next step: whether it scales to large in-the-wild egocentric video such as EgoDex, while further improving IDM accuracy